

# Module 6 Week B — Stretch: Cross-Lingual Embedding Comparison

**Honors Track.** This is an Honors Track stretch assignment. Stretch assignments are for learners who have completed all core assignments, are On Track or Advanced, and are attending consistently. If you are behind on core work, focus there first.

**Due:** Thursday end of day. Resubmissions are accepted through Saturday of the assignment week.

**Prerequisite:** None – this stretch uses base lab embedding skills from Week B.

## The Challenge

Multilingual transformer models like `bert-base-multilingual-cased` are trained on text from 104 languages simultaneously. In theory, they learn a shared embedding space where semantically similar content clusters together regardless of language. But how well does this actually work for Arabic and English climate text?

Investigate cross-lingual embedding quality by extracting multilingual BERT embeddings for paired English/Arabic texts from the Lab 6A climate dataset and measuring whether same-topic texts cluster together across languages.

This directly addresses a critical question for NLP deployment in the Middle East and North Africa (MENA) region: can a single embedding model power bilingual search, classification, and retrieval without separate per-language models?

## What You'll Learn

- Working with multilingual transformer models ( `bert-base-multilingual-cased` )
- Cross-lingual embedding similarity: do translations cluster together?
- Evaluating embedding quality for bilingual applications
- Creating similarity heatmaps for cross-lingual analysis
- Connecting technical capabilities to deployment decisions in multilingual contexts

## Outcome

Your deliverable extends your **Lab 6A** repo ( `m6-l6a-{your-username}` ) on a new branch. The bilingual `climate_articles.csv` you used for NER in Week A is already in place and includes both English and Arabic climate texts (filter by the `language` column: `en` / `ar` ). Lab 6B's BBC News dataset is English-only and cannot be used for cross-lingual analysis. The deliverable contains:

| Artifact                       | Description                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cross-lingual embedding script | Extracts multilingual BERT embeddings for English and Arabic climate texts                                                                              |
| Similarity heatmap             | Visualizes cross-lingual similarity between English and Arabic text pairs                                                                               |
| 2-paragraph analysis           | (a) How well does the multilingual model capture cross-lingual similarity? (b) What does this mean for building bilingual NLP tools in the MENA region? |

### Constraints:

- Use `bert-base-multilingual-cased` from Hugging Face (not the English-only BERT)
- Select at least 10 English texts ( `language == "en"` ) and 10 Arabic texts ( `language == "ar"` ) from `climate_articles.csv`. Where possible, choose texts that cover the same topics across languages (e.g., an English text about IPCC and an Arabic text about IPCC)
- Extract embeddings using the same tokenizer-model-pooling pipeline from the lab (mean pooling over last hidden states)
- Compute a 20x20 cosine similarity matrix (all English-Arabic pairs)
- Create a heatmap visualization with text labels (first 40 characters) on both axes
- Compare cross-lingual similarity to within-language similarity: are same-topic cross-lingual pairs more similar than random within-language pairs?
- The analysis must reference specific text pairs and their similarity scores – not generic claims about multilingual models

## Tips

- Multilingual BERT uses a shared WordPiece vocabulary across all languages. Arabic text is tokenized into subword pieces that may look unfamiliar – this is expected.
- The model download is ~680MB (larger than DistilBERT). Start the download early.
- Cross-lingual similarity scores will typically be lower than within-language scores for the same topics. The interesting question is whether the ranking is preserved: are same-topic cross-lingual pairs ranked higher than different-topic pairs?
- If you want to strengthen your analysis, also extract embeddings from the English-only DistilBERT model for the English texts and compare: does the multilingual model sacrifice English quality for cross-lingual capability?

## Submission

- In your **Lab 6A** repo ( `m6-l6a-{your-username}` ), create a new branch named `stretch-cross-lingual-embeddings` off `main`
- Add your cross-lingual embedding script (e.g., `stretch_cross_lingual.py` ), the heatmap, and your written analysis (e.g., `stretch_analysis.md` ). Lab 6A's `data/climate_articles.csv` is already in the repo — filter by `language == 'en'` and `language == 'ar'` to build your paired set
- Push the branch and **open a separate pull request from `stretch-cross-lingual-embeddings` to `main`** — do not add these commits to your existing Lab 6A PR. Each stretch is a separate PR
- Paste the **stretch PR URL** into TalentLMS -> Module 6 Week B -> Stretch 6B-S2